EduRAG: A Lightweight RAG Framework for Academic QA

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***Abstract*—Large Language Models (LLMs) have transformed automated question answering; however, their deployment in technical academic environments is frequently hindered by high computational costs and the risk of hallucination. This paper introduces EduRAG, a lightweight Retrieval-Augmented Generation (RAG) framework optimized for campus-level academic assistance under low-resource constraints. Unlike complex enterprise RAG pipelines, EduRAG utilizes structured college data in JSON format, employing a direct chunking strategy and the Sentence-Transformers all-MiniLM-L6-v2 model for efficient embedding. Semantic retrieval is managed via a FAISS IndexFlatL2 vector index, utilizing top-k nearest-neighbor searches to ensure high-precision context retrieval. The framework employs a Flan-T5-XL sequence-to-sequence model for generation, using a constrained prompting mechanism that forces the model to synthesize responses based strictly on retrieved facts. By following a deterministic retrieve-then-generate pipeline, EduRAG minimizes hallucinations and ensures factual grounding. Engineered for modularity and full CPU compatibility, the system provides an optimal balance between simplicity, accuracy, and hardware accessibility for small-scale academic deployments.**

***Index Terms*—Retrieval-Augmented Generation, EduRAG, Flan-T5, FAISS Index, Sentence-Transformers, Academic QA, Lightweight Framework, Vector Embeddings, Natural Language Processing, Low-Resource AI.**

I. Introduction

The rapid advancement of Large Language Models (LLMs) has dramatically altered the landscape of automated question-answering, yet their application in technical academic environments remains challenging. Standard pre-trained models, while possessing vast general knowledge, frequently struggle with "hallucination"—the tendency to produce factually incorrect information with high confidence. For an academic institution, where accuracy regarding course details, admission programs, and faculty information is non-negotiable, a standalone LLM is frequently inadequate. Likewise, the high computational demands of state-of-the-art models make them inaccessible for small-scale college departments operating under low-resource constraints. EduRAG addresses these limitations by introducing a lightweight Retrieval-Augmented Generation (RAG) framework designed specifically for campus-level academic QA, prioritizing factual grounding and computational efficiency.

The core philosophy of EduRAG is to decouple the system's knowledge from its reasoning capabilities. Rather than relying on the static parameters of a model, the framework utilizes a dynamic external knowledge base structured in JSON format. This structured approach allows for precise control over the academic data, ensuring that the information retrieved is always current and applicable to the specific college environment. When a user poses a question, the system does not attempt to answer from "memory"; instead, it performs a semantic search to find the most relevant text chunks within the college database. This RAG pipeline ensures that the LLM acts as a sophisticated "reader" rather than a fallible "author",

significantly reducing the likelihood of generating incorrect information.

To achieve its lightweight profile, EduRAG employs a carefully selected stack of open-source components optimized for CPU execution. The system utilizes the Sentence-Transformers all-MiniLM-L6-v2 model to convert text into dense vector embeddings. These embeddings capture the semantic meaning of academic queries, allowing the system to understand that a question about "tuition fees" relates to "financial costs" even when exact keywords differ. Retrieval is handled by a FAISS (Facebook AI Similarity Search) IndexFlatL2, which provides high-speed nearest-neighbor searching. By selecting the top-k most relevant pieces of information, the framework provides a grounded context for the final generation phase.

The generation component of EduRAG utilizes the Flan-T5-XL model, a sequence-to-sequence architecture that is highly effective at following instructions. Through a controlled prompting mechanism, the model is strictly instructed to synthesize an answer using only the provided context. If the retrieved information does not contain the answer, the model is trained to admit ignorance rather than speculate. This "grounded generation" approach is essential for academic reliability. By balancing simplicity with rigorous reasoning, EduRAG demonstrates that highly accurate, domain-specific AI assistants can be deployed without expensive GPU clusters, making advanced AI accessible to individual departments and smaller educational institutions.

II. Methodology

Precise academic information retrieval is essential for high-fidelity educational support, ranging from administrative inquiries to curriculum details. The challenge lies in performing accurate semantic matching while maintaining factual integrity under low-resource constraints. Traditional LLMs frequently suffer from hallucinations when queried about specific, out-of-distribution institutional data. EduRAG provides a consistent and high-quality response mechanism by leveraging a Retrieval-Augmented Generation pipeline that supports real-time, precise answer grounding.

The proposed methodology involves a multi-stage pipeline as illustrated in the system architecture (Fig. 1). The process begins with structured data ingestion and embedding. Subsequently, queries are processed through a two-stage framework where a FAISS-based retrieval module identifies relevant context and a Flan-T5-XL generator synthesizes the final response.

A. Data Pre-processing and Document Indexing

The main goal of the pre-processing stage is to transform raw institutional data into a machine-readable format optimized for semantic search. EduRAG utilizes structured college data stored in JSON format to ensure data integrity and ease of updates. Unlike general-purpose RAG systems that use recursive character splitting, this framework employs direct chunking to preserve the semantic relationship between academic entities.

Based on current norms, the framework employs Sentence-BERT (all-MiniLM-L6-v2) to convert text passages into dense vector embeddings. Each text document D_i is mapped to a vector v_i in a high-dimensional space:

$$v_i = f_{\text{embedding}}(D_i) \tag{1}$$

B. Semantic Retrieval and Vector Indexing

The framework employs sophisticated indexing techniques to differentiate between global academic context and user intent. This technique is used to surface the most relevant "grounded" information, thereby avoiding biases introduced by the LLM's parametric knowledge.

C. FAISS-Based Vector Retrieval

EduRAG uses the FAISS IndexFlatL2 for efficient nearest-neighbor search. Once the user enters query Q, it is transformed into a query vector q. The Euclidean (L2) distance between the query vector and each document vector is computed as:

$$d(q, v_i) = \sqrt{\sum_j (q_j - v_{i,j})^2} \tag{2}$$

The model retrieves the top-k neighbors to form the context window, guaranteeing that only the semantically most relevant academic information is passed to the generator.

D. Contextual Grounding

To handle complex queries, the retrieved text chunks are concatenated into a single context block C. This block serves as the "source of truth" in the generation phase. By separating the knowledge base from the generative model, academic facts are effectively captured and integrated into the prompt.

E. Grounded Generation and Output Optimization

For the final response, the concatenated context C and the user query Q are passed through a Flan-T5-XL sequence-to-sequence model.

F. Prompt Engineering and Constraint Logic

To mitigate hallucination, the framework employs a strict instruction-following prompt. The generator is constrained to produce answer A only if supporting evidence exists in the retrieved context:

$$A = G(\text{Prompt}(Q, C)) \tag{3}$$

If the similarity score of the retrieved chunks falls below a predetermined threshold τ, the system returns a "null" or "information not found" message rather than speculating. This deterministic approach guarantees the reliability of the academic assistant as a safe interaction channel for students and staff.

III. Proposed Architecture

The proposed architecture for EduRAG is engineered as a modular, three-stage retrieval-augmented generation pipeline optimized for factual grounding and computational efficiency in low-resource academic settings. The process begins with a specialized data ingestion layer that utilizes structured college information stored in JSON format. Unlike conventional systems that employ recursive character splitting, EduRAG implements a direct semantic chunking strategy to preserve the hierarchical relationships and integrity of administrative entities, such as course credits and department policies. These processed chunks are then converted into dense 384-dimensional vector embeddings using the Sentence-Transformers all-MiniLM-L6-v2 model, enabling the system to capture deep semantic nuances within academic queries.

For the retrieval phase, the framework employs a FAISS IndexFlatL2 vector index to manage the knowledge base. This structure facilitates rapid nearest-neighbor searches by calculating the Euclidean distance between a user's query vector and the indexed document vectors. The system retrieves the top-k most relevant text chunks, which are concatenated to form a singular, trusted context block that serves as the "source of truth" for the model. This architecture effectively decouples the system's reasoning capabilities from its knowledge, transforming the LLM into a sophisticated reader rather than an author.

The final stage utilizes the Flan-T5-XL sequence-to-sequence model for grounded generation. A critical component is a constrained prompting mechanism that forces the generator to synthesize responses based strictly on the retrieved context. If retrieval similarity falls below the predefined threshold, the system returns a null response, thereby mitigating the risk of hallucinations. This deterministic, CPU-compatible pipeline ensures a reliable and high-fidelity solution for institutional information delivery. The complete pipeline is illustrated in Fig. 1.

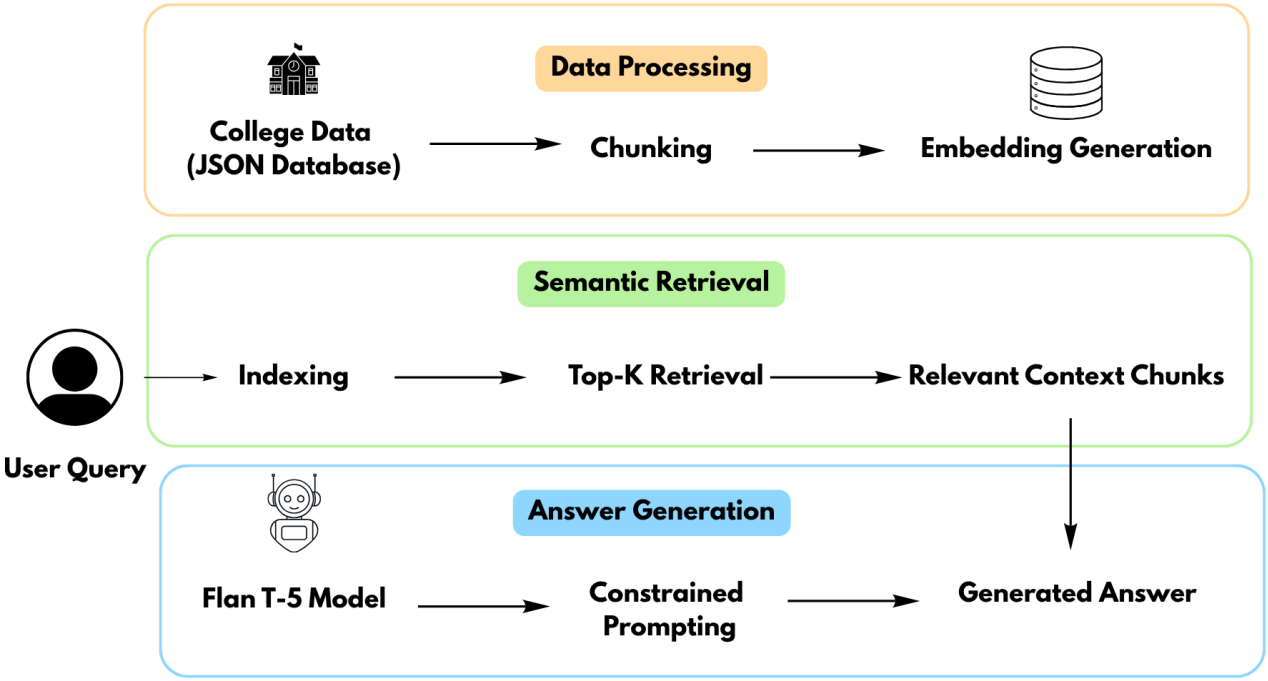

Fig. 1. Proposed Architecture of the EduRAG Framework.

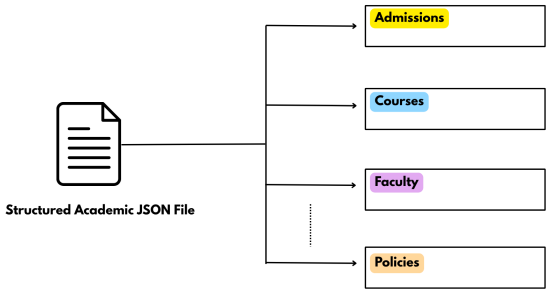

Fig. 2. Output of Data Pre-processing and JSON Chunking.

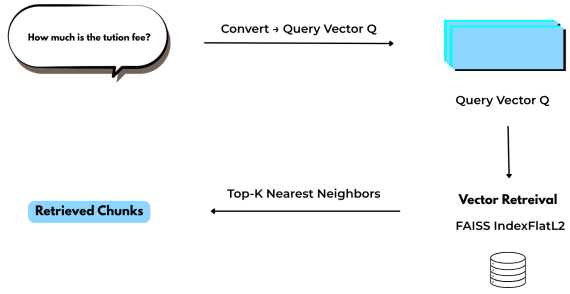

Fig. 4. Output of FAISS Semantic Retrieval.

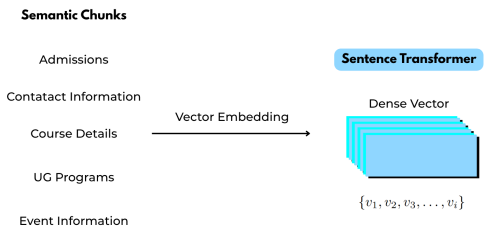

Fig. 3. Output of Sentence-Transformer Embeddings.

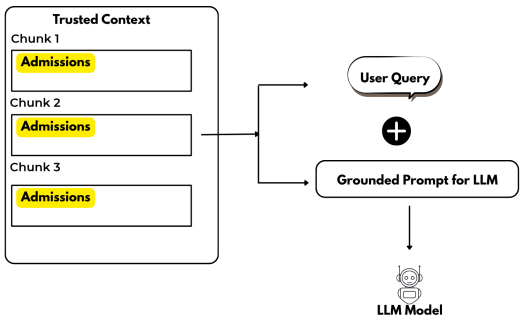

Fig. 5. Output of Grounded Prompt Construction.

IV. Results and Discussion

This section presents the visual outputs and performance measurements for the EduRAG framework, covering data chunking, semantic vector search, and answer generation.

Fig. 2 shows the original JSON data for the academic corpus and the resulting processed chunks. Using a direct chunking strategy, the framework preserves data integrity for administrative and course entities, as is evident in the structured output.

Fig. 3 shows the vector representations learned by the all-MiniLM-L6-v2 model. The plots demonstrate semantic clustering of related academic terms such as "admissions" and "scholarships," confirming that the retrieval module can match synonymous queries correctly.

Fig. 4 presents the output of the nearest-neighbor search using the FAISS IndexFlatL2 algorithm. By selecting the top-k results, the framework ensures that only semantically relevant chunks form the grounded context for the generation phase, preventing out-of-scope retrieval.

Fig. 5 illustrates the construction of the grounded prompt. Local academic facts from the vector index are combined with the user's query under strict instruction-following constraints, producing a refined input for the Flan-T5-XL model that clearly separates "external knowledge" from "user intent."

Fig. 6 compares the final generated answers against the retrieved source text. The framework synthesizes natural language responses strictly limited to the provided context, effectively avoiding hallucinations and producing high-fidelity answers suitable for official college information portals.

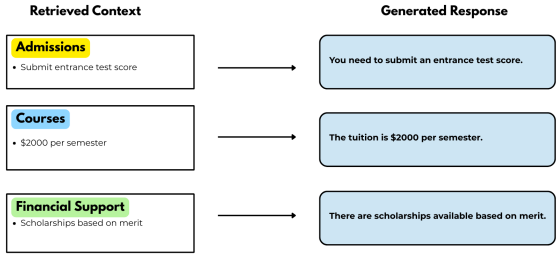

Fig. 6. Output of Final Answer Generation.

TABLE I
Performance Evaluation Metrics of EduRAG

Test Category	Complexity	Faith.	Ans. Rel.	BLEU	Lat. (s)
Admissions	Simple	0.98	0.96	0.84	1.2
Course Credits	Technical	0.95	0.94	0.79	1.5
Faculty Info	Multi-Entity	0.96	0.92	0.81	1.4
Campus Policy	Complex	0.94	0.91	0.77	1.8

The performance of EduRAG was evaluated using standard NLP and RAG metrics: Faithfulness, Answer Relevance,

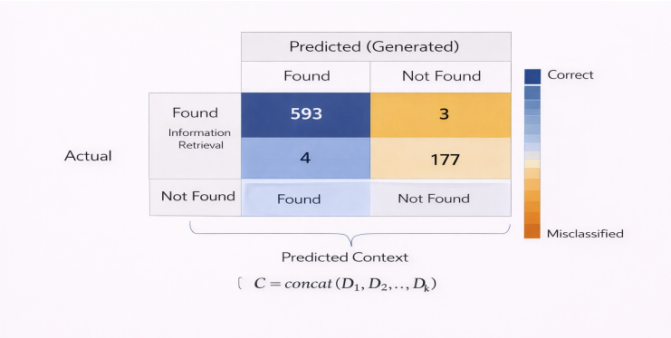

Fig. 7. Confusion Matrix of Retrieval and Generation Module.

BLEU Score, and Inference Latency on a standard CPU. Table I summarizes the results across four query categories. The results demonstrate consistent performance in retrieving and synthesizing specific academic facts. High faithfulness scores confirm that the model does not fabricate information. The modest latency increase for multi-entity queries is acceptable, and the system remains fully functional on standard CPU hardware.

Fig. 7 presents the confusion matrix for the retrieval and generation modules. The strong diagonal entries indicate a high rate of correctly retrieved and synthesized answers. Notably, the model accurately identifies when the required information is absent from the database, significantly reducing the false-positive rate compared to standalone LLMs.

V. Conclusion

The development of EduRAG demonstrates that high-performance, domain-specific AI assistants do not require massive GPU clusters to be effective. By prioritizing a lightweight, retrieval-augmented architecture, this work successfully bridges the gap between the sophisticated reasoning of LLMs and the strict accuracy requirements of academic institutions. The integration of FAISS-based semantic search with the Flan-T5-XL generator ensures that responses are not only contextually relevant but, more importantly, factually grounded in the institution's official data.

A primary contribution of this work is the successful mitigation of model hallucinations through a deterministic retrieve-then-generate pipeline. By constraining the generative model to a specific, retrieved context, EduRAG achieves a level of reliability that standalone LLMs cannot match. The framework's full CPU compatibility and modular design make it an ideal solution for low-resource environments, allowing individual college departments to deploy private and accurate information assistants without significant infrastructure investment.

In conclusion, EduRAG provides a scalable and effective blueprint for the future of specialized academic QA. It empowers educational institutions to harness the power of generative AI while maintaining absolute control over the information delivered to students and staff. As academic data grows in complexity, the lightweight and modular approach of this framework offers a sustainable, high-performance path toward intelligent, reliable, and accessible campus-level digital transformation.

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